Anusha Lavanuru

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Education

Columbia University, New York Expected Dec 2024
- Master of Science, **Computer Science**
- **Courses:** UI Design, ML, NLP, Algorithms, Databases, VR/AR, Computer Graphics, Human Computer Interaction
- **Teaching Assistant:** UI Design (Fall '24), Introduction to Databases(Spring '24), Computing in Context(Fall '23)

Gokaraju Rangaraju Institute of Engineering and Technology, India July 2017 - May 2021
- Bachelor of Technology, **Computer Science and Engineering**
- **Courses:** Algorithms, Data Structures, AI, Advanced Calculus, Probability and Statistics, Big Data, OS, Networks

Skills

Languages :	C, C++, Java, Python, React, JavaScript, TypeScript, SQL, postgresQL, AWS, HTML, CSS
Tools :	SpringBoot, Git, Figma, Scala, Jira, Jenkins, Conference, Unity, Blender, OpenGL
Python Libraries :	OpenCV, Pandas, NumPy, Scikit, matlab, TensorFlow, MySQL, Pytorch, NLTK

Work Experience

Research Assistant | Computer Graphics and User Interfaces Lab Sept 2024 - present
- Developed **React** and **Spring Boot** based full-stack features for a geospatial app, integrating **RESTful APIs** for real-time data visualization
- Optimized UI performance by 30% using **React Hooks**, **lazy loading**, and server-side caching with **Spring Boot**
- Built scalable backend services with **Spring Boot**, **PostgreSQL**, and secure session handling for dynamic data processing
- Enhanced XR environments using **Unity**, **Blender**, and **OpenGL**, enabling seamless interaction between 2D/3D views and back-end systems

Research Assistant | Computer Enabled Abilities Lab May 2024 - present
- Built a full-stack **AR application** using **Unity**, **Blender**, **ReactJS**, and **Spring Boot**, integrating real-time data with **RESTful APIs**
- Designed a scalable **PostgreSQL** database to store dynamic historical data and ensured secure data handling with **Spring Boot's session management**
- Optimized application performance with **React Hooks**, **lazy loading**, and efficient state management, reducing load times by 30%
- Delivered seamless AR experiences by integrating 3D assets with real-time data rendering and back-end services for interactive user engagement

SDE Intern | NomadFi May 2024 - Sept 2024
- Built modular **React** components for a fintech platform, including the homepage, dashboard, and settings
- Utilized **React Hooks, Context, and Redux** for advanced state management, enhancing user experience
- Integrated **Spring Boot** APIs to optimize performance, reducing data fetch latency by 25%
- Developed an **OCR system** in **Python** and **Spring Boot**, automating invoice processing and structuring data in an **SQL database**

Software Engineer | Shure Audio Technologies Aug 2021 - Aug 2023
- Designed and implemented scalable software features for the **Designer NextGen** platform using **Python**, **Flask**, and **PostgreSQL**, enhancing system reliability by 25% and supporting 35+ new functionalities
- Automated 100+ test cases by building reusable frameworks with **Selenium**, **Pytest**, and **CI/CD pipelines**, reducing manual QA time by 70% and ensuring seamless deployments
- Developed backend services for licensing systems and integrated them with a **React** front-end, enabling a 40% increase in software sales and Shure's first $1 billion revenue milestone
- Refactored legacy codebases to improve performance by 30%, incorporating **RESTful APIs**, **Redis** for caching, and containerized deployments with **Docker** and **Kubernetes**

Projects

NomadFi Sept 2024
- Led a team of **3** to build a **full-stack asset tokenization platform** with **ReactJS, Redux, Material-UI**, and a **Spring Boot** backend using **PostgreSQL** and **RESTful APIs**, reducing data processing time by 25%

FitBlend June 2024
- Developed a user-friendly **UI** application called Fit Blend using **ReactJS** to help users explore and understand smoothies, incorporating interactive features and real-time data integration for enhanced user experience